A

PROJECT REPORT

On

"Micro Algae based Air Purifier"

Submitted to

MIT ADT University, Pune

in partial fulfillment of the requirements for the Mini-Project of the course

Computer Engineering Workshop (23BTCS1107)

Submitted by

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FIRST YEAR - ASH | **A.Y 2023-24** | **SEMESTER-II**

CERTIFICATE

This is to certify that the Project report entitled
"**Micro Algae based Air Purifier**"

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is a bonafide work carried out by them, under the supervision of Dr. Pranav Chippalkatti and it is submitted towards the partial fulfillment of the requirement of MIT Art, Design and Technological University, Pune for the Mini-Project of Computer Engineering Workshop.

Dr. Pranav Chippalkatti
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MIT ADT University

Place: Pune

Date: 10/05/2024

FIRST YEAR - ASH **A.Y 2023-24** **SEMESTER-II**

DECLARATION

I hereby declare that the Project entitled "**Micro Algae based Air Purifier**" submitted towards the partial fulfillment of the requirement of MIT-ADT University, Pune for the Mini-Project of Computer Engineering Workshop is a record of bonafide work carried out by us under the supervision of Dr. Pranav Chippalkatti, Assistant Professor, Department of Electrical and Electronics Engineering (Computer Science Engineering), MIT School of Computing, Pune.

We further declare that the work reported in this report has not been submitted and will not be submitted, either in part or in full, for any other work in this institute or any other institute or university.

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ABSTRACT

Air pollution is a significant concern in Indian cities, causing various health problems. Current air pollution control measures are not entirely effective, and access to clean air is scarce. However, there is a promising solution that uses eco-friendly technologies to purify the air. The proposed solution is a microalgae-based outdoor air purifier that can consume pollutants in the air and release clean oxygen. This purifier is made of a transparent fibre plastic tank filled with water and microalgae to maintain its effectiveness in purifying the air. The air is pumped into the tank via pipes with holes at a regulated rate. An ARDUINO circuit is used to pump air. With support from the government and other institutions, this technology has the potential to provide an eco-friendly and sustainable solution to the problem of air pollution in Indian cities.

Keywords:

1. Air pollution
2. Microalgae-based outdoor air purifier
3. Eco-friendly technology
4. Sustainable solution
5. Residential Areas

PHOTOGRAPH OF THE PROJECT

Fig: Prototype of Micro Algae based Air Purifier

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Chapter 1
INTRODUCTION

1.1 Introduction

Air pollution possess a great threat to urban residencies in India, driven by rapid urbanization and industrialization. Traditional purification methods have proven inadequate, necessitating innovative alternatives. Our project introduces a novel approach: harnessing the natural air-purifying abilities of microalgae.

Microalgae, found abundantly in aquatic environments, can absorb pollutants like carbon dioxide and nitrogen oxides while releasing oxygen through photosynthesis. Our micro algae-based air purifier system cultivates microalgae within a transparent tank, providing a sustainable solution to air pollution.

With low energy consumption and minimal maintenance, our microalgae-based approach offers promise for combating air pollution in Indian residential areas. This report explores the scientific basis, technical implementation, and potential impact of our eco-friendly solution, aiming to catalyse broader efforts toward sustainable urban development and environmental stewardship.

1.2 Organization of the Report

Chapter 1: Introduction
This section provides an overview of the project, highlighting the critical air pollution issue in residential areas and introducing our proposed microalgae-based solution.
Chapter 2: Background and Rationale
We explore the causes and impacts of air pollution, alongside existing purification methods, laying the groundwork for our innovative approach.
Chapter 3: Project Objectives
We define clear goals and targets for our project, outlining how we aim to address air pollution through microalgae-based purification.
Chapter 4: Methodology
Details our approach to designing and implementing the microalgae-based purification system, covering microalgae strain selection, system design, and sensor integration.

Chapter 5: Results and Findings

Presents the outcomes of our experiments and field tests, including data on air quality improvements and the effectiveness of our purification system in real-world conditions.

1.3 Closure

In conclusion, our project represents a significant step forward in the quest for sustainable solutions to air pollution in Indian residential areas. By harnessing the natural purifying capabilities of microalgae, we have demonstrated the potential to create cleaner, healthier environments for urban residents while reducing the carbon footprint of traditional air purification methods. Moving forward, we remain committed to further refining and advancing this technology, working collaboratively with government agencies, research institutions, and community stakeholders to realize its full potential and create lasting positive change for generations to come.

Chapter 2
LITERATURE REVIEW

2.1 Introduction

The literature review section serves as a critical foundation for understanding the current state of knowledge and research surrounding air pollution and air purification technologies. By synthesizing and analysing existing studies, this review aims to provide insights and context for the development of innovative solutions to address air pollution challenges in Indian residential areas.

2.2 Review of Literature

The body of literature related to air pollution in residential areas and air purification technologies is extensive and multifaceted. Studies consistently highlight the severe and escalating impacts of air pollution on public health, environmental quality, and socioeconomic well-being. Research has documented elevated levels of pollutants such as particulate matter (PM), nitrogen oxides (NOx), sulphur dioxide (SO2), and volatile organic compounds (VOCs) in urban areas, stemming primarily from industrial activities, vehicular emissions, biomass burning, and construction.
In response to these challenges, a wide range of air purification technologies have been explored and developed, spanning from traditional methods like mechanical filtration and electrostatic precipitation to more advanced approaches such as photocatalysis, activated carbon adsorption, and nanotechnology-based solutions. While these technologies have shown varying degrees of effectiveness in reducing air pollution, they often face limitations in terms of cost, energy consumption, scalability, and environmental impact.

One promising avenue that has garnered increasing attention in recent years is the use of biological agents, particularly microalgae, for air purification. Microalgae, owing to their photosynthetic properties and ability to sequester pollutants while releasing oxygen, offer a natural and sustainable means of improving air quality. Research has demonstrated the efficacy of microalgae-based systems in removing pollutants such as CO_2, NO_x, SO_2, and VOCs from the air, with the added benefit of biomass production for potential secondary applications.

2.3 Summary of Literature Review

Synthesizing the findings from the reviewed literature, it becomes evident that air pollution represents a pressing and multifaceted challenge in residential areas, with significant implications for public health, environmental quality, and socioeconomic well-being. The prevalence of pollutants such as particulate matter, nitrogen oxides, sulphur dioxide, and volatile organic compounds underscores the urgent need for effective air purification strategies. While traditional air purification methods have been explored extensively, they often exhibit limitations in terms of efficiency, cost-effectiveness, and environmental impact. Emerging technologies offer promise in addressing these shortcomings, with biological approaches, particularly microalgae-based systems, gaining attention for their natural and sustainable means of improving air quality.

Microalgae, leveraging their photosynthetic properties, have demonstrated the ability to sequester pollutants from the air while releasing oxygen, making them an attractive candidate for air purification applications. Research has shown the effectiveness of microalgae-based systems in removing pollutants such as carbon dioxide, nitrogen oxides, sulphur dioxide, and volatile organic compounds, offering potential benefits beyond air purification, including biomass production and wastewater treatment.

However, despite the promise of microalgae technology, there remains a gap in the literature and practical implementation tailored to the unique context of Indian urban environments. Comprehensive research and development initiatives focusing on optimizing microalgae-based air purification systems for Indian cities are lacking. Addressing this gap presents an opportunity to develop innovative and sustainable solutions to mitigate air pollution and enhance environmental quality in Indian urban areas.

2.4 Problem Statement

Despite advancements in air purification technologies, the problem of air pollution persists unabated in Indian cities, necessitating novel and sustainable solutions. Current methods often fall short in effectively reducing pollutant levels while maintaining environmental integrity and affordability. Moreover, there is a lack of comprehensive research and development initiatives focusing specifically on microalgae-based air purification systems tailored to the unique context of Indian urban environments. Thus, there exists a critical gap in the literature and practical application of microalgae technology for air purification in India, which our project aims to address.

2.5 Objectives

1. Develop a microalgae-based air purification system optimized for Indian urban settings.
2. Evaluate the effectiveness of the microalgae-based system in reducing air pollution levels.
3. Assess the system's impact on environmental sustainability and public health.
4. Investigate the scalability and feasibility of implementing microalgae-based air purification technology in diverse urban contexts.
5. Generate genuine data and insights to contribute to the body of knowledge on sustainable air purification solutions for Indian cities.

2.6 Closure

In conclusion, the literature review underscores the urgency and importance of addressing air pollution in Indian cities through innovative and sustainable means. By leveraging insights from existing research, our project seeks to contribute to this endeavour by pioneering a microalgae-based air purification solution tailored to the specific needs and challenges of Indian urban environments. Through interdisciplinary collaboration and scientific inquiry, we aim to advance knowledge and technology in this critical area and make meaningful contributions to improving air quality and enhancing the well-being of urban populations.

Chapter 3

METHODOLOGY

3.1 Introduction

This chapter outlines the methodology employed in the development of the microalgae-based outdoor air purifier. It discusses the proposed steps to be executed in order to bring the project to reality, including the research, design, and implementation phases. Additionally, it delves into the technical details of the purifier's functionalities and the various components used to make it operational.

3.2 Methodology details

Our methodology for designing and developing the microalgae-based outdoor air purifier involved several technical aspects, including creating a 3D model, manufacturing, calibration, assembly, and programming. We combined these processes to create an efficient and sustainable air purifier that effectively purifies the air.

3.2.1 To study the literature for project

A comprehensive study of relevant literature has been conducted to evaluate the feasibility and effectiveness of the proposed microalgae-based outdoor air purifier. The literature review indicates that air pollution is a significant threat to public health worldwide and has a severe impact on the quality of life, particularly in urban areas. The use of microalgae for air purification has been identified as a promising and eco-friendly solution to this problem.

Several studies have demonstrated that microalgae can effectively remove pollutants from the air, including nitrogen oxides, sulphur dioxide, carbon monoxide, and volatile organic compounds, among others. Chlorella sp. and Scenedesmus sp. have been identified as suitable species of microalgae for air purification due to their high photosynthetic activity and efficient pollutant removal capabilities.

3.2.2 Selection of Material

The selection of materials for the microalgae-based air purifier was a critical factor in ensuring its efficiency and durability. The transparent plastic tank was chosen for its lightweight and easy handling, allowing for better light penetration and easy detection of microalgae growth. Rubber pipes were selected for their flexibility and small holes, enabling smooth air flow. High-precision sensors such as the PMS5003 particulate sensor, DHT22 humidity and temperature sensor, MHZ19 CO2 sensor, and BH1750 photodetector were chosen for accurate air quality monitoring. Lastly, a Mega2560 microcontroller was selected for its power, reliability, and quick data processing.

3.3 Closure

The study's first step is to design and develop a microcontroller-based circuit that includes a Mega2560 microcontroller, a display, a particulate sensor, a humidity and temperature sensor, a CO2 sensor, and a photodetector. The circuit will be programmed to display air quality data and control the air pump to regulate air flow in the tank based on processed air quality data.

The second step is to select suitable microalgae species that are efficient in consuming pollutants from the air and releasing clean oxygen. Based on research, Chlorella sp. and Scenedesmus sp. are the most suitable microalgae species for air purification in the proposed purifier. Additional nutrition like sodium bicarbonate will be provided for maintaining the purifier's effectiveness.

The third step is to integrate the electronic components and the selected microalgae species into the purifier's design. The purifier's design will consist of a transparent plastic tank, a rubber pipe with small holes throughout placed at the base of the tank, and a PVC pipe connected to an air pump that will regulate air flow in the tank.

The fourth step is to test the purifier's performance by measuring the air quality data with sensors and comparing it with predetermined air quality standards. The purifier's effectiveness in reducing pollutants in the air and releasing clean oxygen will also be evaluated.

CHAPTER 4

DESIGN AND SOFTWARE ANALYSIS

4.1 Introduction

The design and software analysis section provide insights into the conceptualization and technical aspects of the microalgae-based air purifier. It outlines the design process, including rough conceptual sketches, and discusses the software analysis conducted to ensure the effectiveness and efficiency of the proposed solution.

4.2 Design of the Robot

The design of the air purifier involved creating a blueprint to address various challenges, including purification efficiency, maintenance, and durability. Using tools like AutoCAD, the design was divided into three main parts: the tank, the piping system. Detailed illustrations, including CAD models and schematics, are presented to visualize the design.

Fig 4.1: Rear View of the Air Purifier

Fig 4.2: Side View of the Air Purifier

4.3 Analysis of Air purifier

The analysis phase involved assessing the structural integrity and performance of the air purifier. key components, such as the tank, to ensure resilience under operational conditions. Results, efficiency of reducing pollution.

4.4 Circuit of the Air Purifier

The electronic components and circuitry of the air purifier are outlined, highlighting the use of Arduino as the main controller and also for pump control. Specifications of components and their connections are detailed, along with a block diagram illustrating the system's architecture.

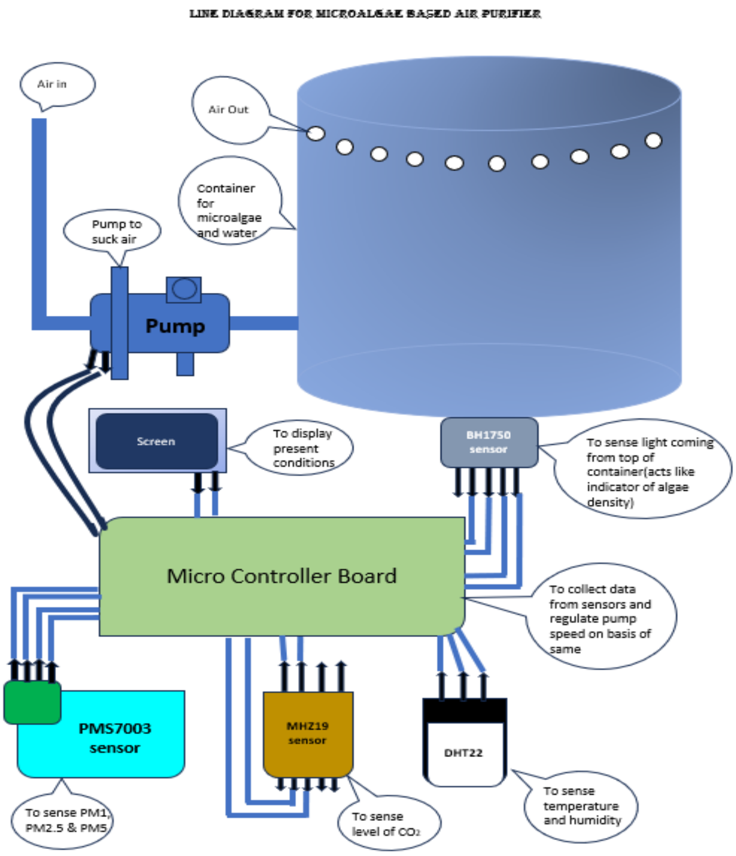

Fig 4.3 : Block Diagram of the Air Purifier

4.5 Testing

The testing phase involves validating the mechanical and electronic functionalities of the air purifier. Various tests, including sensor calibration and system integration, are conducted to ensure proper operation and performance.

4.6 Operational Workflow

The operational workflow of the air purifier is described step by step, outlining the procedures for setup, calibration, and deployment. Detailed instructions are provided for configuring and controlling the system, ensuring seamless operation in real-world scenarios.

1. Data is collected by sensors and passed to microcontroller.
2. On the basis of data collected calculations are performed.
3. On the values of calculations pump is turned on or off.
4. This makes air purifier work effectively.

4.7 Closure

In conclusion, the design and software analysis section explain the comprehensive approach undertaken to develop and evaluate the microalgae-based air purifier. By integrating advanced design principles and software simulations, the proposed solution aims to provide an effective and sustainable means of addressing air pollution challenges in Indian residential areas.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 Introduction

This chapter presents the results of the project work conducted with the microalgae-based air purifier. Key findings and discussions regarding the effectiveness, design, and potential future developments of the purifier are elaborated upon.

5.2 Key points in Results and Discussion

For a project utilizing microalgae-based purifiers, consider integrating an Arduino Mega 2560 microcontroller along with various sensors such as light sensors, CO_2, CO, SO_2 detectors, and other relevant sensors. With this setup, you can construct a small-scale system designed to clean water or air using microalgae effectively. This involves creating a controlled environment where microalgae can thrive and filter out pollutants. The Arduino Mega 2560 can be programmed to monitor environmental parameters such as light intensity, CO_2 levels, and pollutant concentrations detected by the sensors. By analysing this data, you can optimize the growth conditions for the microalgae and assess their efficiency in purifying the water or air. Ultimately, the project aims to demonstrate how microalgae, when coupled with advanced sensing and control technologies like Arduino, can play a vital role in improving environmental health and sustainability by making our surroundings cleaner and healthier.

The design of the purifier, created using AUTO-CAD software, incorporates modern technology features essential for efficient air purification. Detailed analysis and discussions highlight the innovative aspects of the design and its alignment with the project objectives

5.3 Future Scopes

For a microalgae-based purifier project, start by researching different microalgae types and how they clean the surrounding air. Set up a small system to grow them, maybe in containers with water and nutrients. Then, introduce polluted water or air and see how well the microalgae clean it over time. Adjust the setup as needed and observe results carefully to improve the process. Finally, document what you find, see how effective your purifier is, and think about how it could help with real environmental problems. The goal is to show how microalgae can make our environment cleaner and safer.

5.4 Closure

This section summarizes the various results obtained from the project work, including the merits and limitations of the microalgae-based outdoor air purifier. Additionally, future opportunities and advancements in cleaning technology are discussed, emphasizing the potential for continued innovation and development in the field of sustainable air purification solutions.

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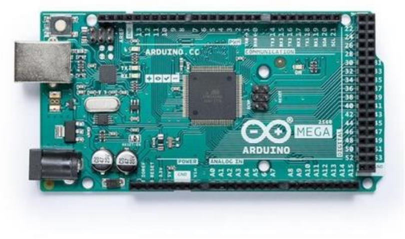
APPENDIX I

ARDUINO MEGA 2560

1. Microcontroller: It uses the ATmega2560 microcontroller.
2. Operating Voltage: 5V.
3. Input Voltage (recommended): 7-12V.
4. Input Voltage (limits): 6-20V.
5. Digital I/O Pins: 54 (of which 15 provide PWM output).
6. Analog Input Pins: 16.
7. DC Current per I/O Pin: 20 mA.
8. DC Current for 3.3V Pin: 50 mA.
9. Flash Memory: 256 KB, of which 8 KB is used by the bootloader.
10. SRAM: 8 KB.
11. EEPROM: 4 KB.
12. Clock Speed: 16 MHz.
13. Length: 101.52 mm.
14. Width: 53.3 mm.
15. Weight: 37 g.
16. USB Interface: ATmega16U2 (on the rev3 version).
17. Power Supply: Can be powered via USB connection or an external power supply.
18. Reset Button Included for resetting the microcontroller.
19. Operating Temperature: 0°C to 85°C.
20. Development Environment: Compatible with the Arduino IDE and other development environments supporting Arduino programming.



❖Sensor Working Principle
Name : PMS7003

Working principle:
The PMS7003 operates on the principle of laser scattering. It emits a laser beam into the air sample and detects the light scattered by particles passing through the beam. By analyzing the scattered light, it determines the concentration of particles in different size ranges, such as PM1.0, PM2.5, and PM10.

Specifications:
- Particle Size Range: PM1.0, PM2.5, PM10
- Measurement Principle: Laser scattering
- Measurement Range: 0.3 to 1.0 µm for PM1.0, 0.3 to 2.5 µm for PM2.5, 0.3 to 10 µm for PM10
- Measurement Accuracy: ±10%
- Output: Concentration in µg/m³ (micrograms per cubic meter)
- Response Time: ≤10 seconds
- Operating Voltage: 5V DC
- Operating Temperature Range: -10°C to 60°C
- Operating Humidity Range: 0% to 99% RH (non-condensing)
- Dimensions: 50mm x 46mm x 17mm
- Weight: Approximately 25 grams

❖Sensor Working Principle
Name : MHZ19
Working Principle:
The MH-Z19 sensor operates based on the principle of NDIR spectroscopy. It contains an infrared light source, a gas sample chamber, and an infrared detector. When the infrared light passes through the gas sample chamber, it interacts with CO2 molecules, absorbing specific wavelengths of light. The amount of light absorbed is proportional to the concentration of CO2 in the sample. The infrared detector measures the intensity of light that passes through the chamber and calculates the CO2 concentration based on the absorption of infrared light by CO2 molecules.
Specifications:
- Measurement Range: 0 to 5000 ppm (parts per million)
- Measurement Principle: Nondispersive infrared (NDIR) spectroscopy
- Accuracy: ±50 ppm ±3% of reading
- Resolution: 1 ppm
- Response Time: <60 seconds
- Warm-up Time: <3 minutes
- Operating Voltage: 3.6V to 5.5V DC
- Operating Current: 18mA (average), 35mA (peak)
- Operating Temperature Range: 0°C to 50°C
- Operating Humidity Range: 0% to 95% RH (non-condensing)
- Dimensions: 33mm x 20mm x 8mm
- Weight: Approximately 5 grams

❖Sensor Working Principle

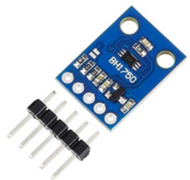
Name : DHT22

Working Pricniple:
The DHT22 sensor measures temperature and humidity using a capacitive humidity sensor and a thermistor. The humidity sensor changes its capacitance based on air moisture, converted to digital signals by an analog-to-digital converter. The thermistor detects temperature changes through resistance variation. The sensor's microcontroller processes these signals to output digital temperature and humidity readings.

Specifications:
- Temperature Measurement Range: -40°C to 80°C
- Temperature Measurement Accuracy: ±0.5°C
- Humidity Measurement Range: 0% to 100% RH (Relative Humidity)
- Humidity Measurement Accuracy: ±2% RH
- Resolution: 0.1°C for temperature, 0.1% RH for humidity
- Operating Voltage: 3.3V to 5.5V DC
- Operating Current: 0.3mA (standby), 2.5mA (active)
- Sampling Interval: 2 seconds (minimum)
- Response Time: 2 seconds (typical) for both temperature and humidity
- Output: Digital signal (single-wire serial interface)
- Dimensions: 15.1mm x 25mm x 7.7mm
- Weight: Approximately 3 grams

❖Sensor Working Principle

Name : BH1750



Working Principle:
The BH1750 sensor operates based on the principle of light intensity measurement using a photodiode. It contains a photodiode that converts light energy into electrical current. The sensor's internal circuitry amplifies and converts this current into a digital signal. The intensity of the digital signal is proportional to the amount of light falling on the photodiode. By measuring this digital signal, the sensor can determine the ambient light intensity.

Specifications:
- Measurement Range: 0 to 65535 lux
- Measurement Resolution: 1 lux
- Measurement Accuracy: ±20% (50 to 10000 lux)
- Operating Voltage: 2.4V to 3.6V DC
- Operating Current: 0.12mA (typical)
- Response Time: 16ms (typical)
- I2C Address: 0x23 or 0x5C (selectable)
- Operating Temperature Range: -40°C to 85°C
- Dimensions: 2.4mm x 2.4mm x 0.8mm (typical)
- Weight: Approximately 0.04 grams

❖Indicator

Name : 0.96' OLED

Working Principle:
An OLED display consists of thin organic layers sandwiched between two electrodes, namely the anode and the cathode. When an electric current is applied, electrons are injected from the cathode and holes from the anode into the organic layers. These electrons and holes combine in the emissive layer, causing organic compounds to emit light. The emitted light passes through the transparent anode and cathode, resulting in the display of images or text.

Specifications:
- Screen Size: 0.9 inches (diagonal)
- Resolution: Typically 128x32 pixels or 80x160 pixels
- Display Technology: OLED (Organic Light-Emitting Diode)
- Color Depth: Monochrome (typically white or blue)
- Interface: Typically SPI (Serial Peripheral Interface) or I2C (Inter-Integrated Circuit)
- Operating Voltage: Typically 3.3V or 5V
- Viewing Angle: Wide viewing angle, typically >160 degrees
- Operating Temperature Range: Typically -40°C to 80°C
- Dimensions: Varies depending on the specific model
- Weight: Lightweight, typically a few grams

❖Actuator

Name : Pump

Working principle:

A 5-volt pump works by converting electrical energy into mechanical energy to move fluids. It operates using a small motor and an impeller or propeller to push fluid through its outlet.

Specifications:

- Voltage: 5 volts
- Flow Rate: Volume of fluid moved per unit time (e.g., liters per hour)
- Head: Maximum vertical lift height
- Operating Temperature Range: Suitable temperature range for operation
- Materials: Construction materials, especially those in contact with fluid
- Lifespan: Expected operational lifespan before maintenance or replacement.

ANNEXURE I

```
#include <Wire.h>
#include <SoftwareSerial.h>
#include "DHT22.h"
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>

#define SCREEN_WIDTH 128
#define SCREEN_HEIGHT 64
#define OLED_RESET -1
Adafruit_SSD1306 display(SCREEN_WIDTH, SCREEN_HEIGHT, &Wire,
OLED_RESET);

#define DHTPIN 2
#define DHTTYPE DHT22

DHT22 dht(DHTPIN);

SoftwareSerial co2Serial(18, 19);  // RX, TX for MH-Z19

const int pumpPin = 7;  // Pin connected to the pump
int co2Threshold = 5;   // Change this value according to your
requirement
int pwmpin = 10;

void setup() {
  Wire.begin();
  Serial.begin(9600);
  co2Serial.begin(9600);
  pinMode(pumpPin, OUTPUT);
  pinMode(pwmpin, INPUT);

  // Initialize OLED display
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("SSD1306 allocation failed"));
    for (;;)
      ;
  }
  display.display();  // Display splash screen
  delay(2000);
  display.clearDisplay();
}

void loop() {
  // Read CO2 level from MH-Z19
  int co2Level = readCO2();
```

```
  // Read temperature and humidity from DHT22
  float temperature = dht.getTemperature();
  float humidity = dht.getHumidity();

  digitalWrite(pumpPin, HIGH);

  // Print temperature and humidity on serial monitor and OLED display
  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.print(" °C, Humidity: ");
  Serial.print(humidity);
  Serial.println(" %");
  display.setCursor(0, 0);
  display.print("Temperature: ");
  display.print(temperature);
  display.println("C");
  display.print("Humidity: ");
  display.print(humidity);
  display.println("%");

  // Check CO2 level and control pump
  Serial.print("CO2 level: ");
  Serial.print(co2Level);
  Serial.print("\n");
  Serial.print("\n");
  if (co2Level > co2Threshold) {
    Serial.println("CO2 level is high! Turning on the pump...");
    digitalWrite(pumpPin, HIGH);  // Turn on the pump
    display.println("CO2 level is high! Turning on the pump...");
  } else {
    digitalWrite(pumpPin, LOW);  // Turn off the pump
    display.println("CO2 level is normal.");
  }

  display.display();
  delay(5000);  // Delay between readings
}

int readCO2() {
  int pwmValue = pulseIn(pwmpin, HIGH, 250000); // Read PWM signal with
timeout of 250ms
  if (pwmValue > 0) {
    // Convert PWM signal to CO2 level (assuming linear relationship)
    int co2Level = map(pwmValue, 200, 3200, 0, 5000); // Adjust range
as needed
```

```
    return co2Level;
  } else {
    return 10; // Error reading CO2 level
  }
}
```